Supplemental Files

Supplemental file 1: Original study protocol

Review title
Probiotics for the prevention of antibiotic-associated diarrhoea: A systematic review and meta-analysis

Anticipated or actual start date
08/08/2019

Anticipated completion date
30/05/2021

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Funding sources/sponsors
NA

Conflicts of interest
None known

Collaborators
NA

Review question(s)
(1) Do probiotics reduce the incidence of antibiotic-associated diarrhoea in adults?
(2) Is there a dose-response?
(3) Is there a species-specific response?

Searches
Databases to be searched: CINAHL Plus, EMBASE, MEDLINE (Ovid), Web of Science and Google Scholar from inception to May 2021, as well as a manual search of clinical trial registries and the reference lists of relevant systematic reviews. Additional details about the search strategy can be found in the attached full search strategy.

Condition or domain being studied

Antibiotics are one of the most commonly prescribed medications. They disturb the normal gastrointestinal microbiota, which can lead to diarrhoea. Probiotics are live microorganisms that, when administered in adequate amounts, confer a health benefit on the host. Our objective was to evaluate existing evidence for the use of probiotics in preventing antibioticassociated diarrhoea (AAD) in adults.

Participants/population

Inclusion: Studies of participants 18 years or older who were receiving any antibiotic therapy for any reason
Exclusion: Studies with participants younger than 18

Intervention(s), exposure(s)

Probiotics - any species/strain, dose or formulation

Comparator(s)/control

Placebo, alternative dose, alternative probiotic strain, or no treatment control

Outcome measure(s)

Incidence of diarrhoea as defined by authors during the antibiotic treatment or follow-up phases
Studies were excluded if they examined probiotics only for treatment, rather than prevention, of AAD

Types of study to be included

Randomised controlled trials (RCTs) only
Written in English

Context

Any setting: inpatient or outpatient

Data extraction (selection and coding)

Data will be extracted from each article by two of three of the authors (CG, GK, KL), using a standardised data extraction form following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. This will be housed within Covidence online software (Covidence systematic review software, Veritas Health Innovation, Melbourne, Australia. Available at www.covidence.org). Data items collected to be included: number of patients allocated to each group, probiotic species, formulation, dose and regime, whether a placebo or 'nothing' control was used, and outcome data on incidence of AAD.

Risk of bias (quality) assessment

Risk of bias will assessed using the Cochrane Risk of Bias [24] tool within the Covidence online software. Assessment will be conducted by two of three authors (CG, GK, KL) and discrepancies resolved by consensus.

Strategy for data synthesis

For data analysis we will use the RevMan 5.4.1 statistical software (Review Manager (RevMan) [Computer program]. Version 5.4. The Cochrane Collaboration, 2020.) We will use either a fixed-effects or random-effects model, reported as either odds ratios or risk ratios with corresponding 95% confidence intervals (95% CI). The number needed to treat (NNT) will also calculated for the complete case analysis.

Analysis of subgroups or subsets

Our a priori clinical heterogeneity variables will include probiotic dose and probiotic species.

Type and method of review

Systematic review and meta-analysis

Supplemental file 2: Search strategy

Electronic sources

- CINAHL Plus
- EMBASE
- MEDLINE (Ovid)
- Web of Science
- Google Scholar
- Manual search of grey literature: ○ Reference lists of relevant systematic reviews
    - Clinical trial registries (https://clinicaltrials.gov/ and http://www.anzctr.org.au/)

Boolean keywords / search terms

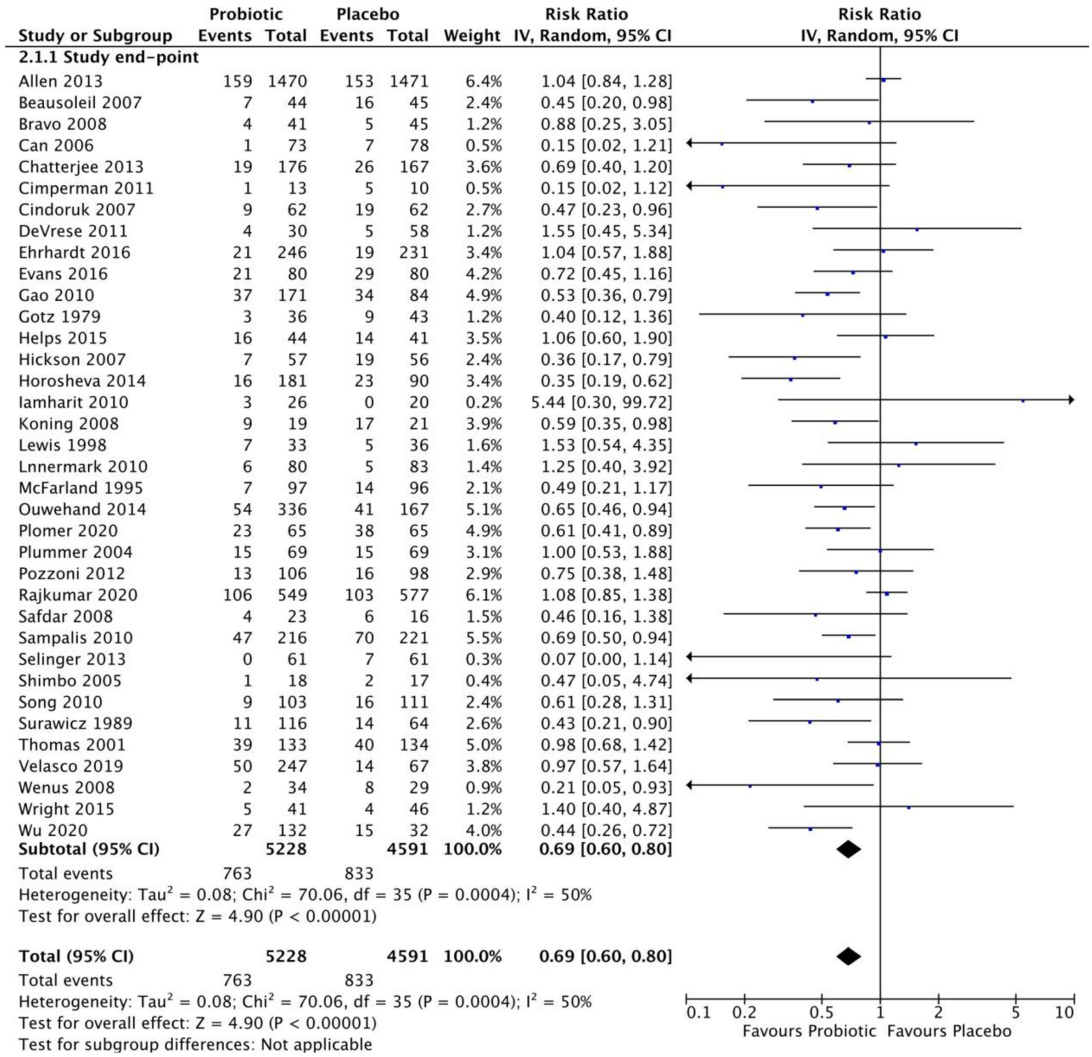
Population
(Adults or humans or aged or elderly or young adults)

Intervention
(Probiotic or fermented milk or fermented yoghurt or lactic acid bacteria or lactobacillus or bifidobacterium or saccharomyces or lactococcus or enterococcus or streptococcus or pediococcus or leuconostoc or bacillus or esherichia or coli or rhamnosus or acidophilus or bifidum or lactis or bulgaricus or reuteri or casei or thermophilus or boulardii or coagulans or delbrueckii or animalis or thermophilus). m_titl.

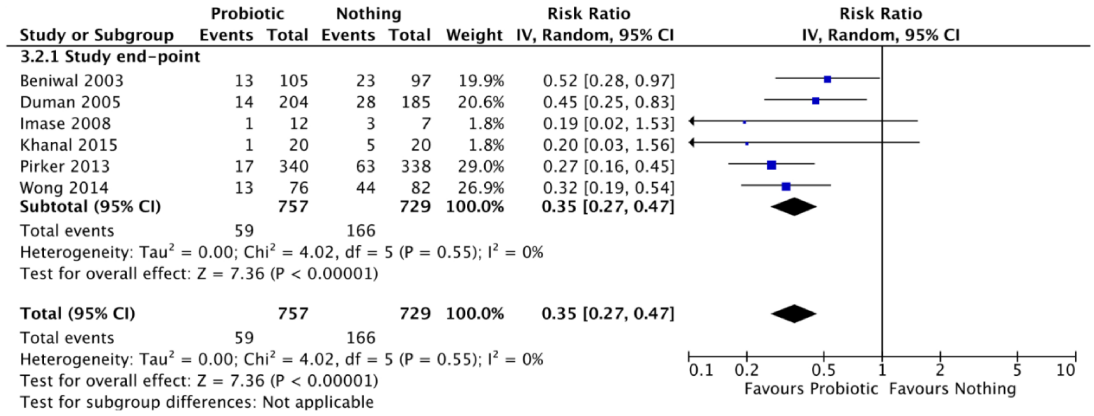
Control
(Placebo or no treatment or pasteurised yoghurt or heat-treated yoghurt or control or compar*). m_titl.

Outcome
(Incidence or prevalence or occurrence or reduction or improvement or prevention or alleviation or episode* or occasion) AND (Diarrh* or antibiotic-associated diarrh* or clostridium difficile associated diarrh* or pseudomembranous colitis or nosocomial diarrh* or infectious diarrh*). m_titl.

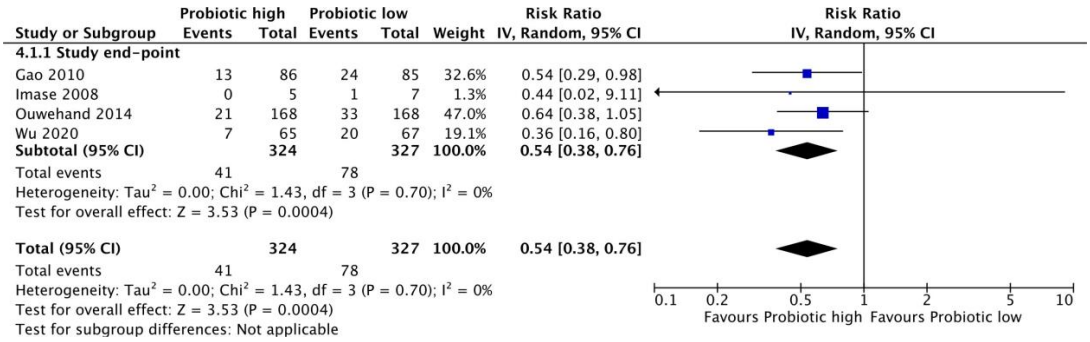
Supplemental file 3: Forest plots of all sub-group analyses



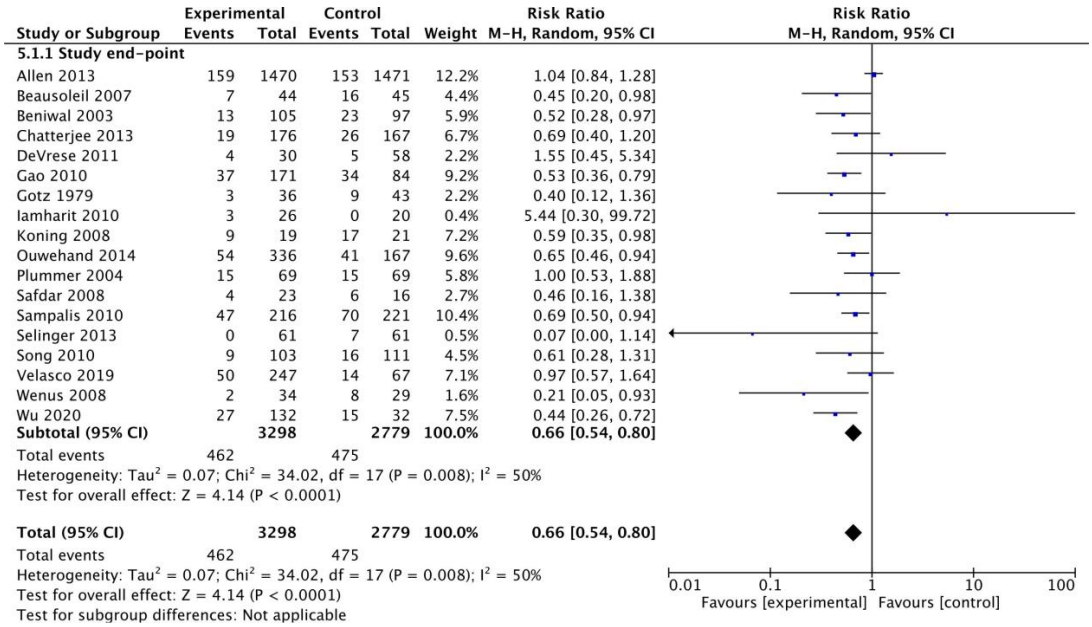
Supplemental figure 1: Subgroup analysis: Placebo effect: Probiotics vs placebo control



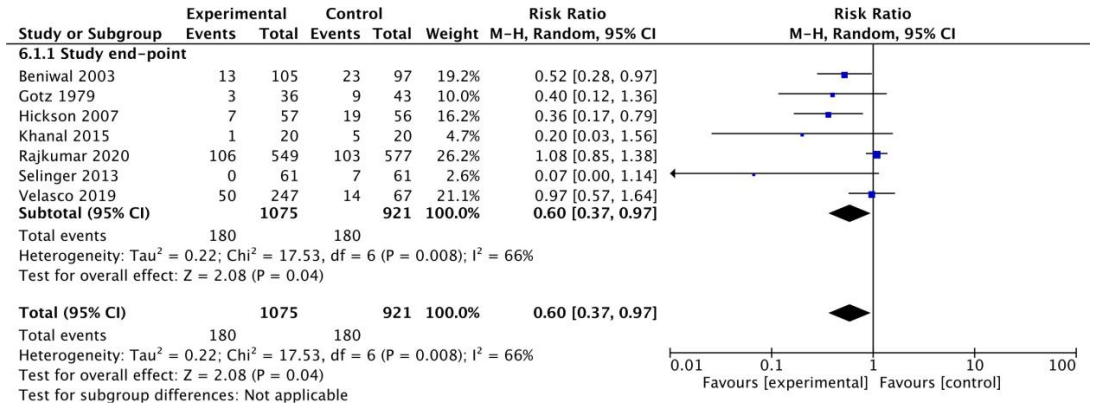
Supplemental figure 2: Subgroup analysis: Placebo effect: Probiotics vs no treatment control



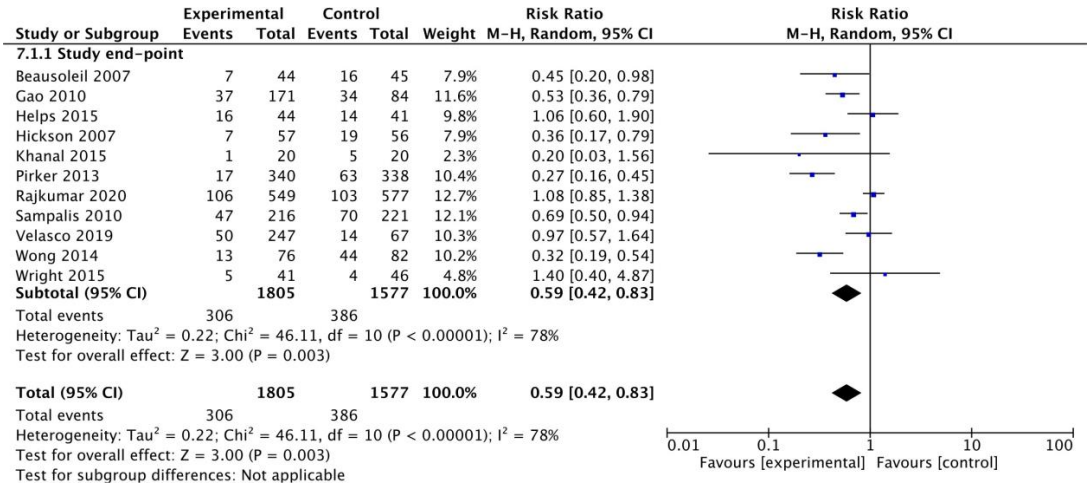
Supplemental figure 3: Subgroup analysis: dose response: High dose vs low dose



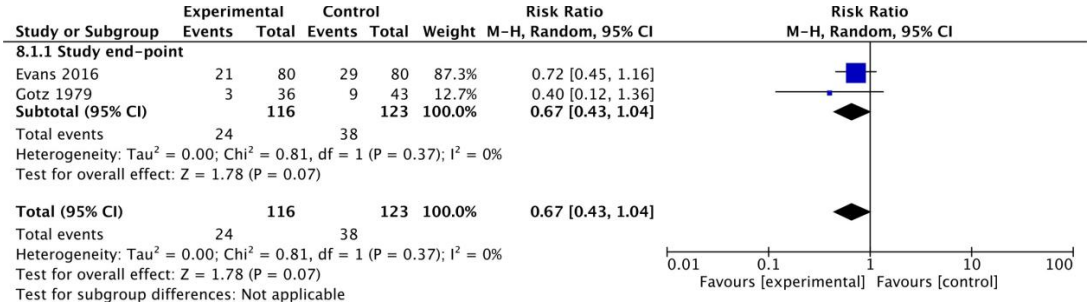
Supplemental figure 4: Subgroup analysis: species: L. acidophilus vs any control



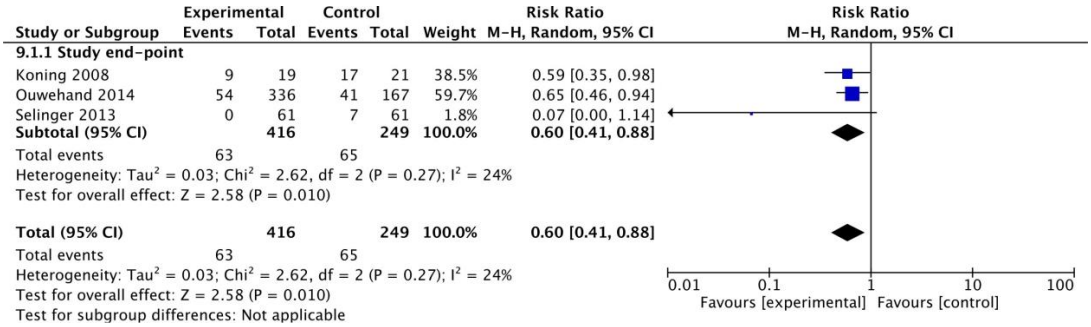
Supplemental figure 5: Subgroup analysis: species: L. bulgaricus vs any control



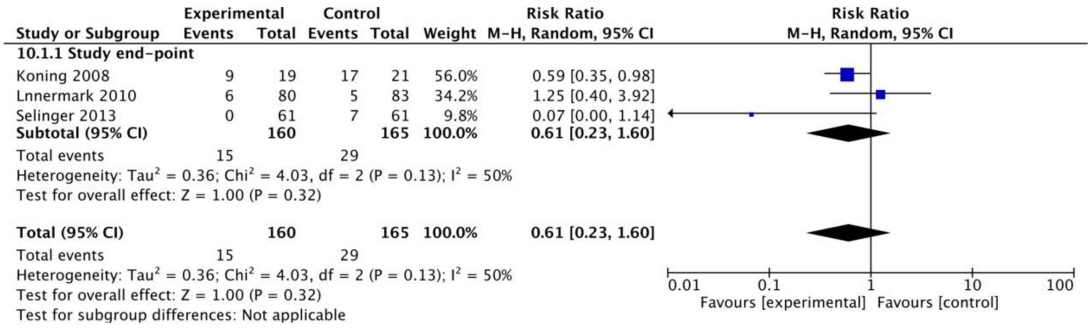
Supplemental figure 6: Subgroup analysis: species: L. casei vs any control



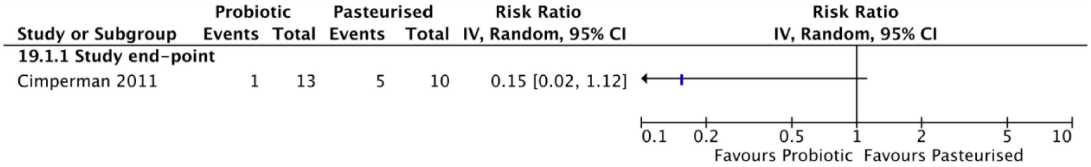
Supplemental figure 7: Subgroup analysis: species: L. helveticus vs any control



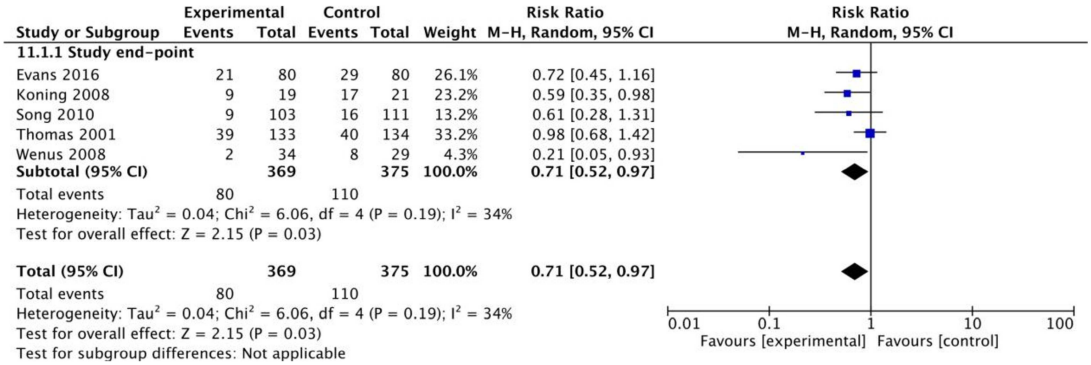
Supplemental figure 8: Subgroup analysis: species: L. paracasei vs any control



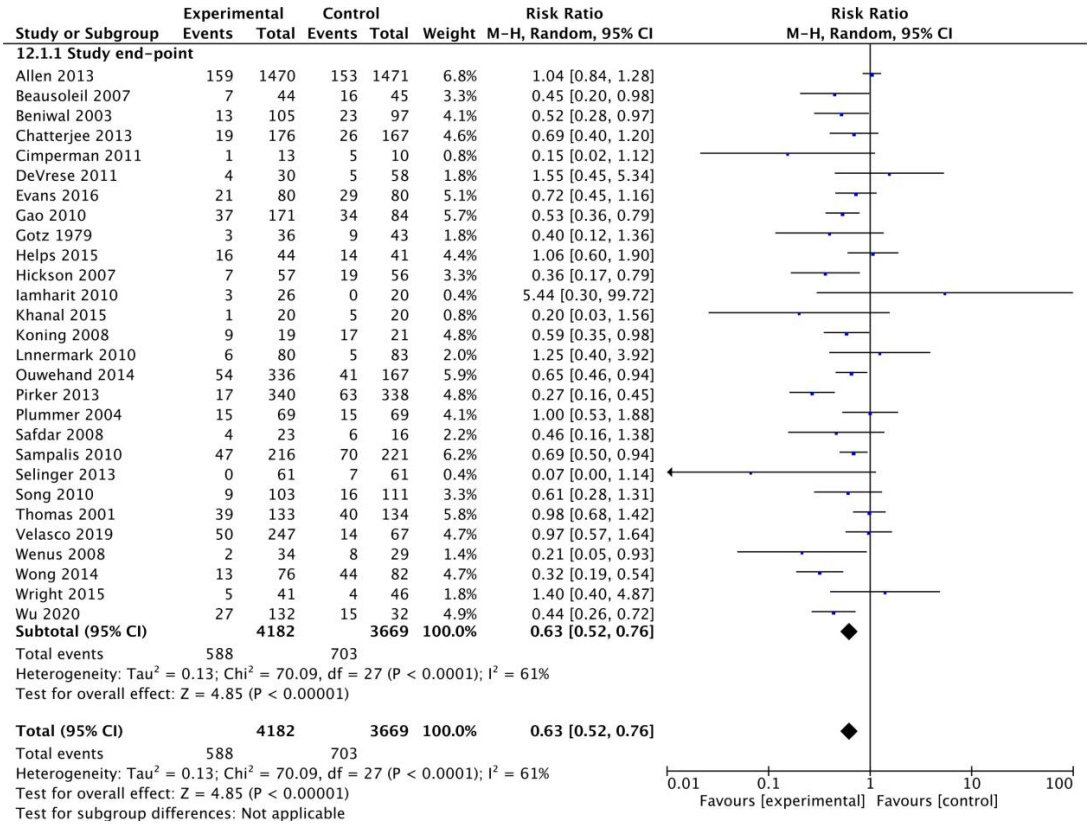
Supplemental figure 9: Subgroup analysis: species: L. plantarum vs any control



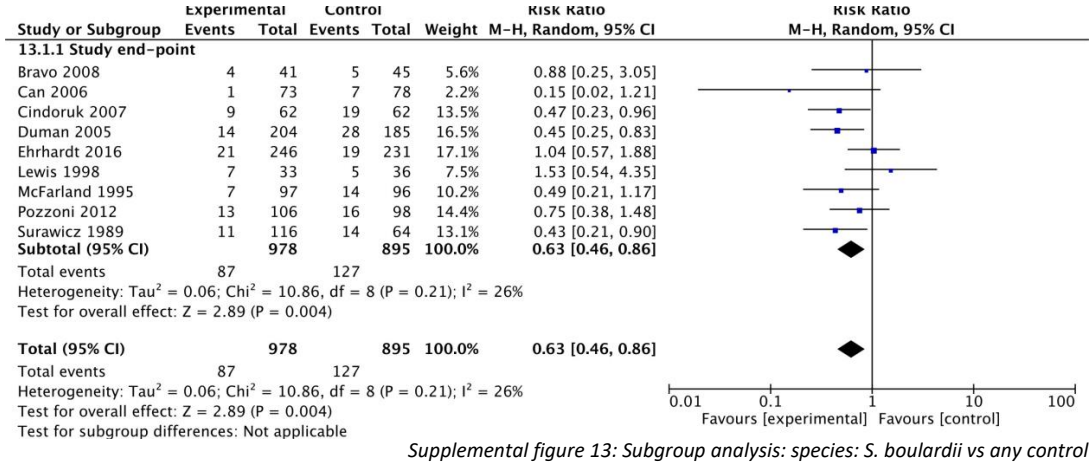
Supplemental figure 10: Subgroup analysis: species: L. reuteri vs any control



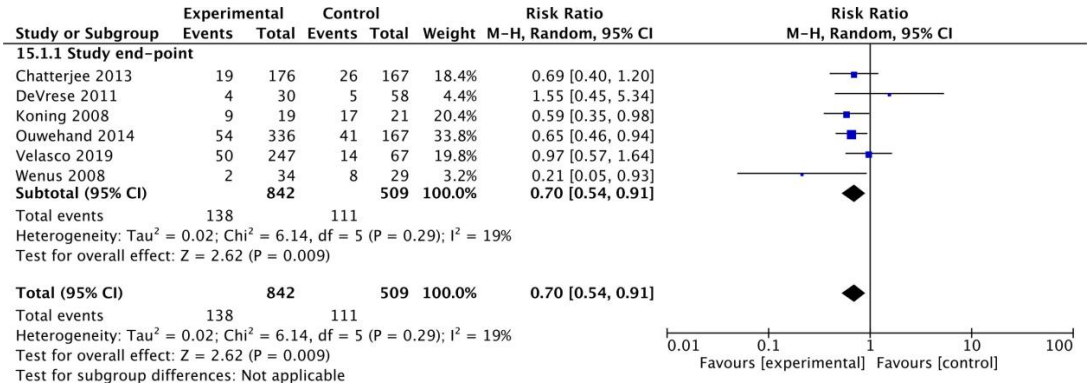
Supplemental figure 11: Subgroup analysis: species: L. rhamnosus vs any control



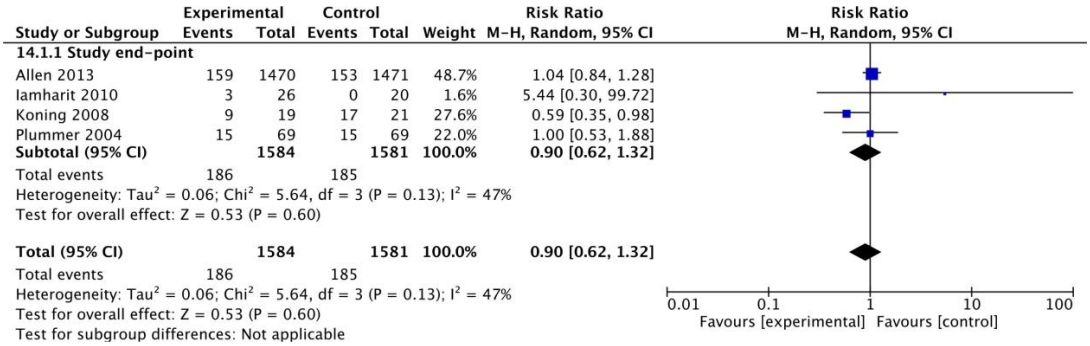
Supplemental figure 12: Subgroup analysis: species: Lactobacillus sp. vs any control



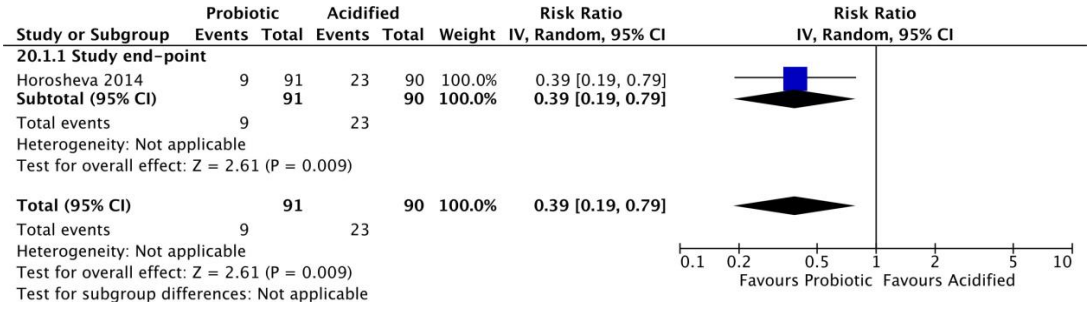
Supplemental figure 13: Subgroup analysis: species: S. boulardii vs any control



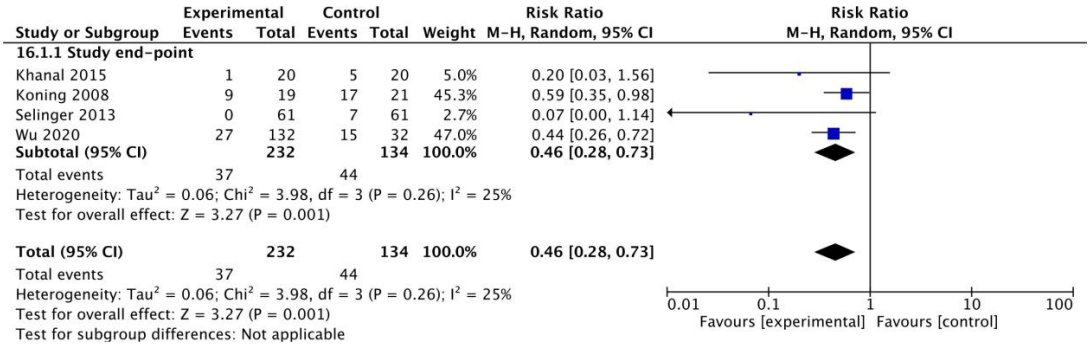
Supplemental figure 14: Subgroup analysis: species: B. animalis ssp. Lactis vs any control



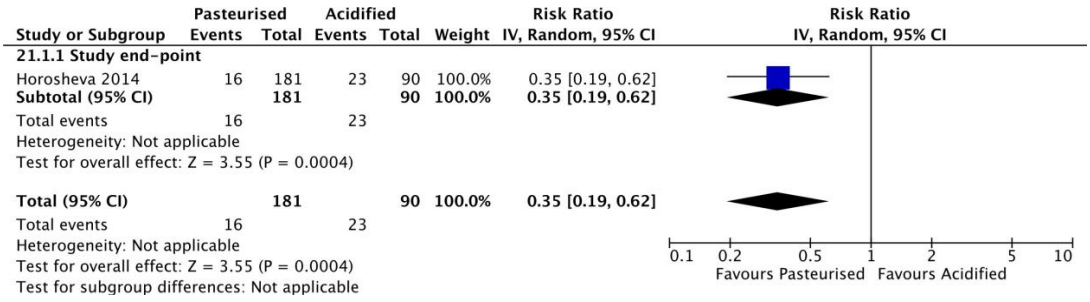
Supplemental figure 15: Subgroup analysis: species: B. bifidum vs any control



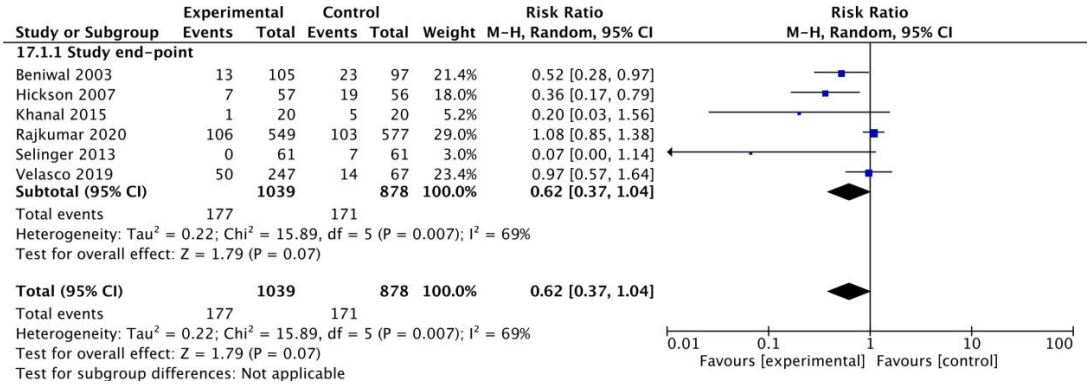
Supplemental figure 16: Subgroup analysis: species: B. licheniformis vs any control



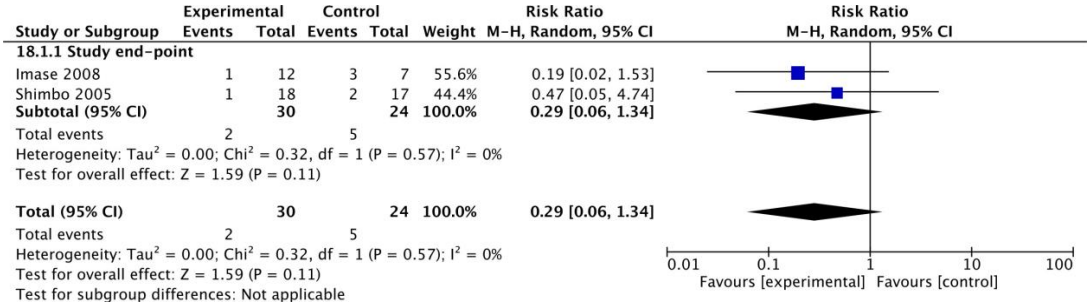
Supplemental figure 17: Subgroup analysis: species: B. longum vs any control



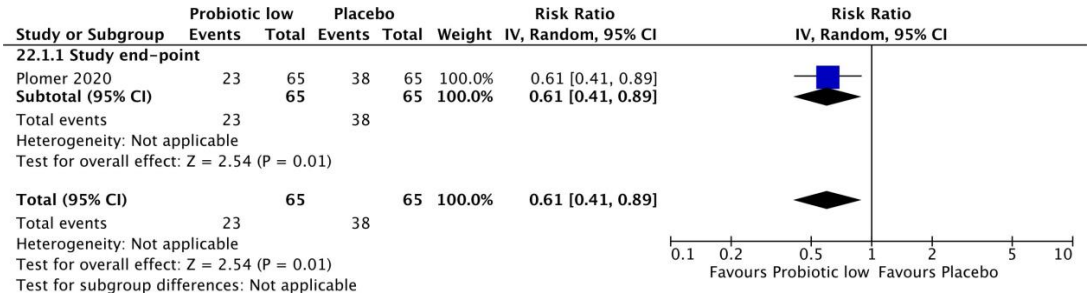
Supplemental figure 18: Subgroup analysis: species: B. subtilis vs any control



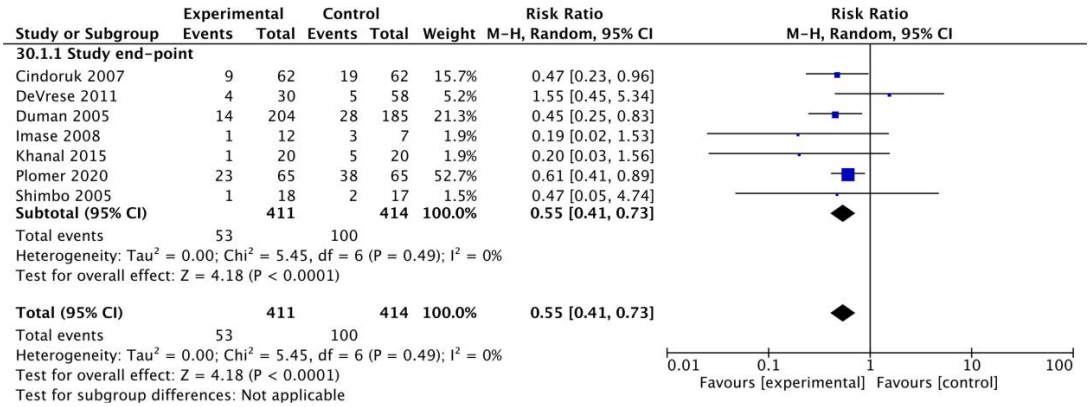
Supplemental figure 19: Subgroup analysis: species: S. thermophilus vs any control



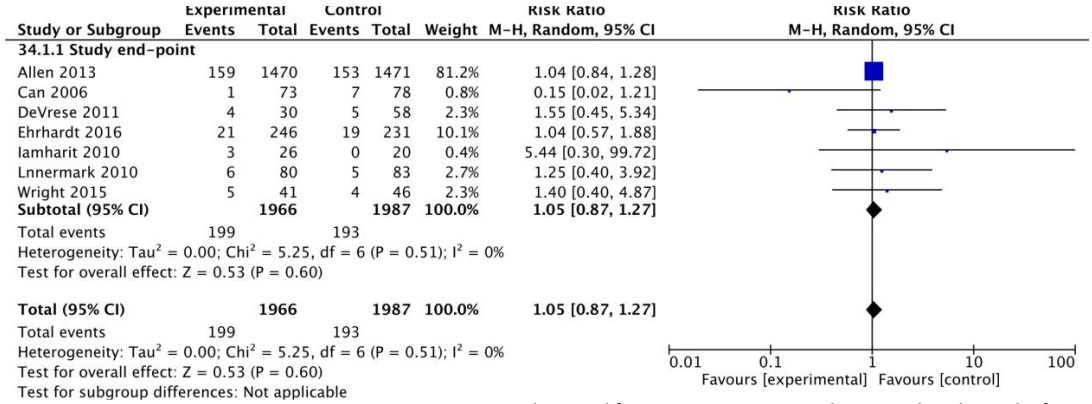
Supplemental figure 20: Subgroup analysis: species: C. butyricum vs any control



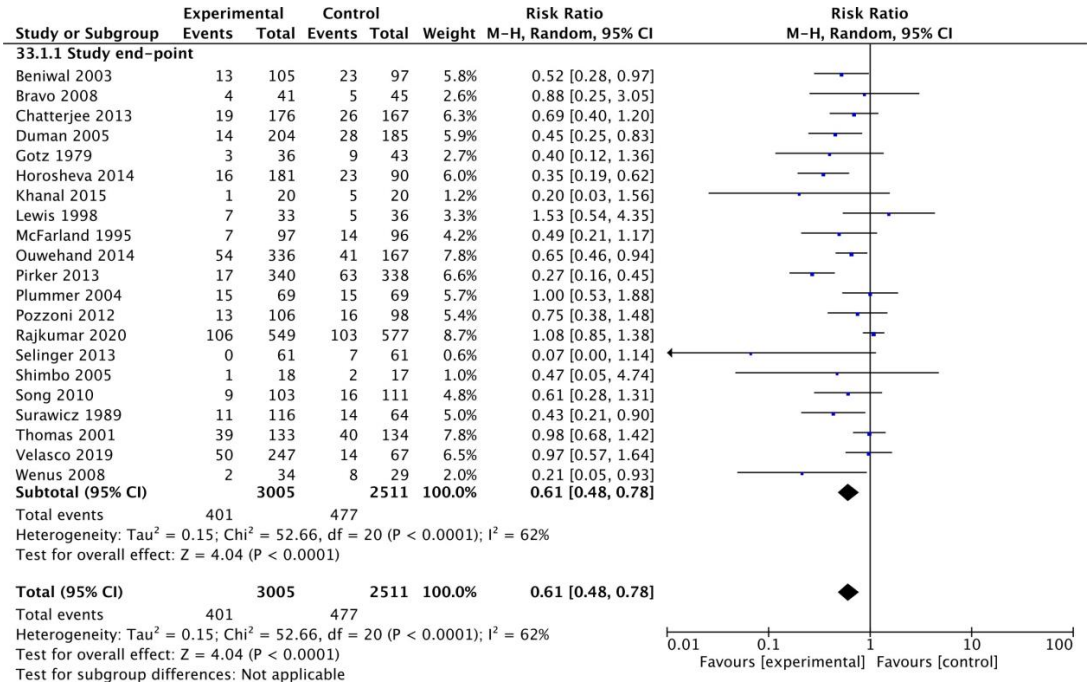
Supplemental figure 21: Subgroup analysis: species: Bacillus clausii vs any control



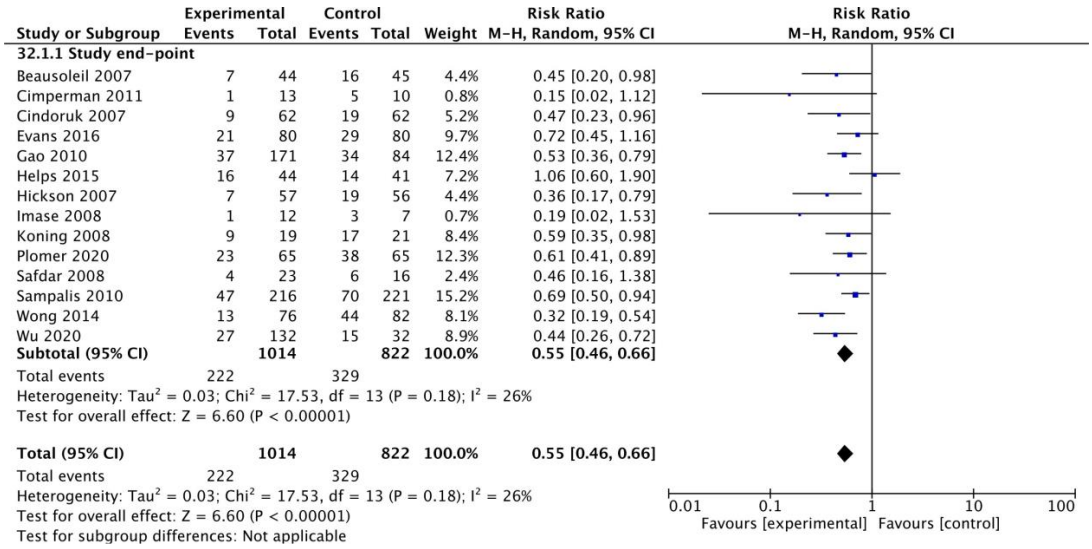
Supplemental figure 22: Subgroup analysis: Probiotics administered concurrently with H. pylori eradication therapy



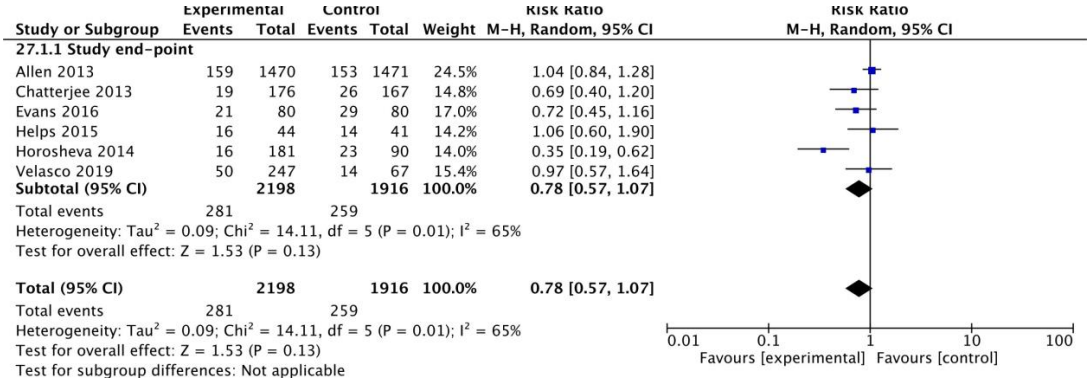
Supplemental figure 23: Sensitivity analysis: Low baseline risk of AAD



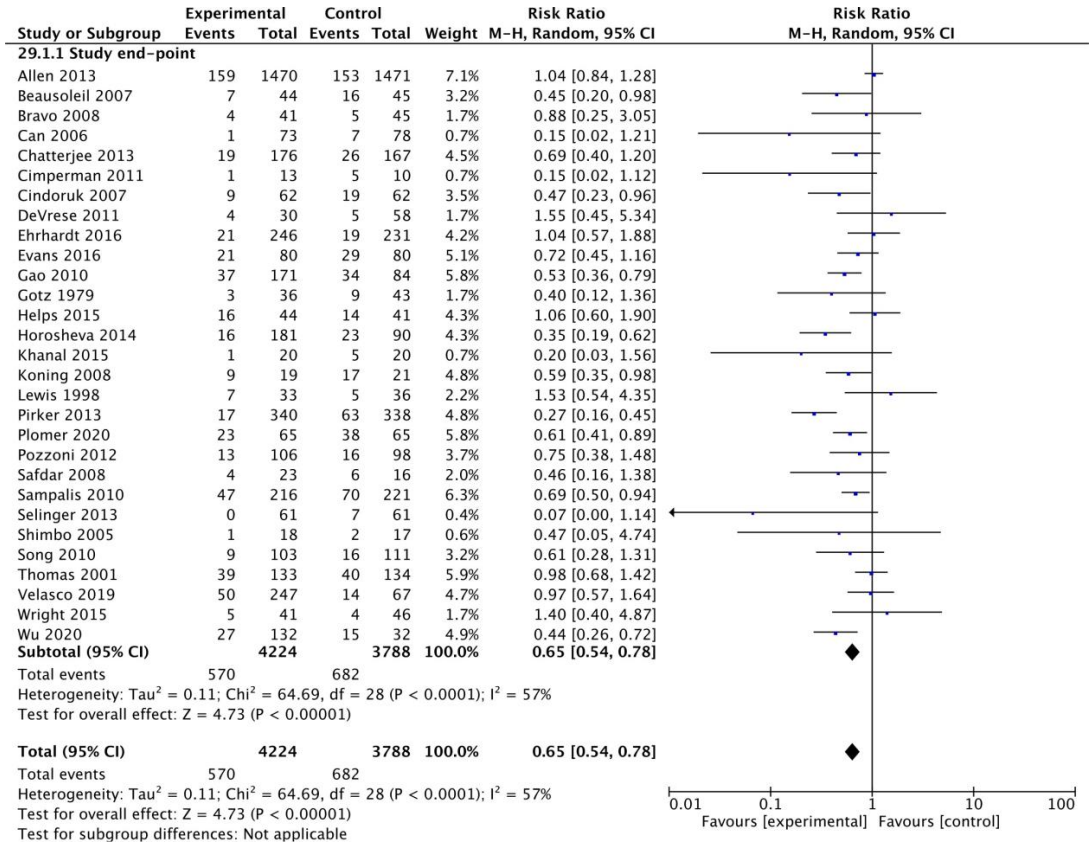
Supplemental figure 24: Sensitivity analysis: Moderate baseline risk of AAD



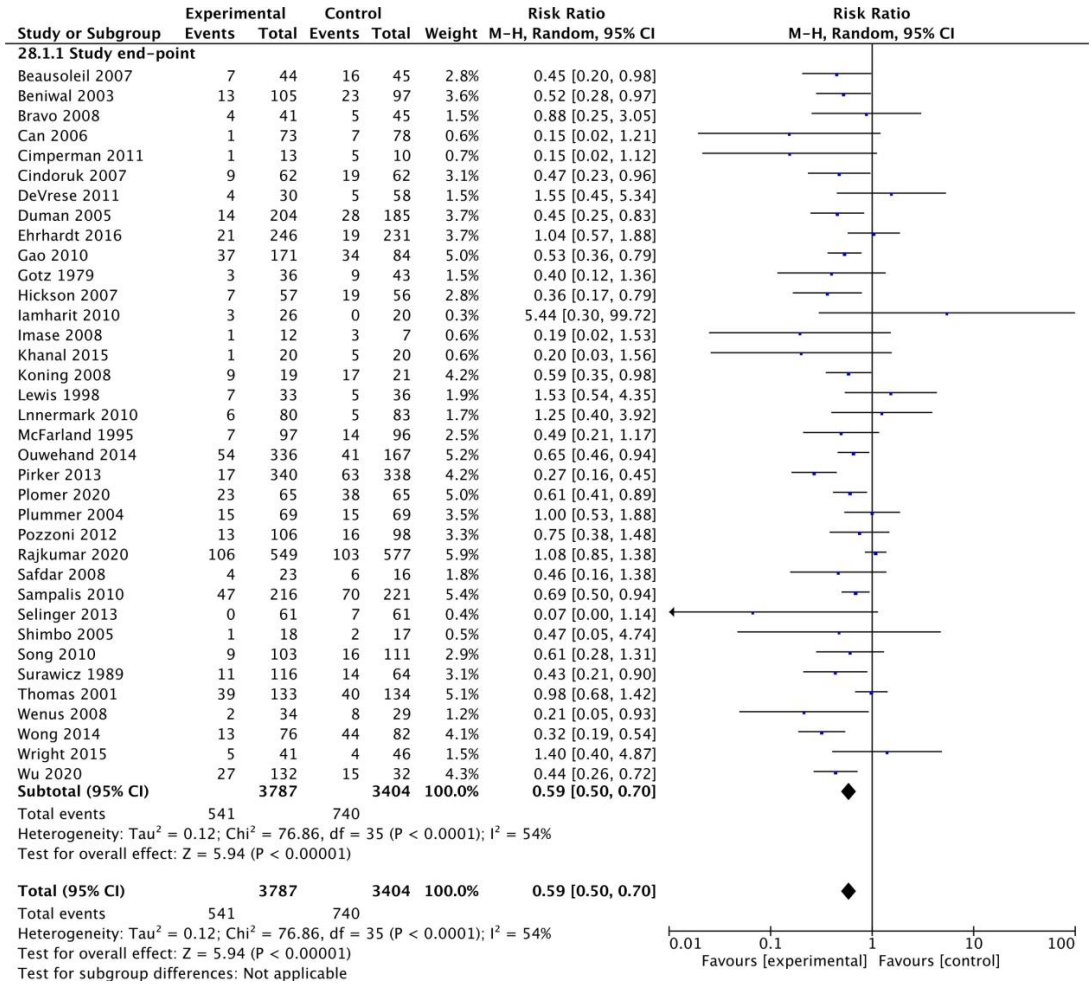
Supplemental figure 25: Sensitivity analysis: High baseline risk of AAD



Supplemental figure 26: Sensitivity analysis: Low risk of bias studies



Supplemental figure 27: Sensitivity analysis: Low/unknown risk of bias studies



Supplemental figure 28: High/unknown risk of bias studies